

MindEval

Benchmarking Language Models on Multi-turn Mental Health Support

José Pombal

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Who I am and what I do



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MindEval is one of the (many) evaluation resources we use to help us build
Dawn, Sword's mental health chatbot

Mental health is a global problem

One billion people globally live with mental health conditions.

AI can be part of a scalable solution

But models are sycophantic, don't set adequate boundaries and reinforce maladaptive beliefs

To improve we need
proper evaluation

Existing benchmarks don't work

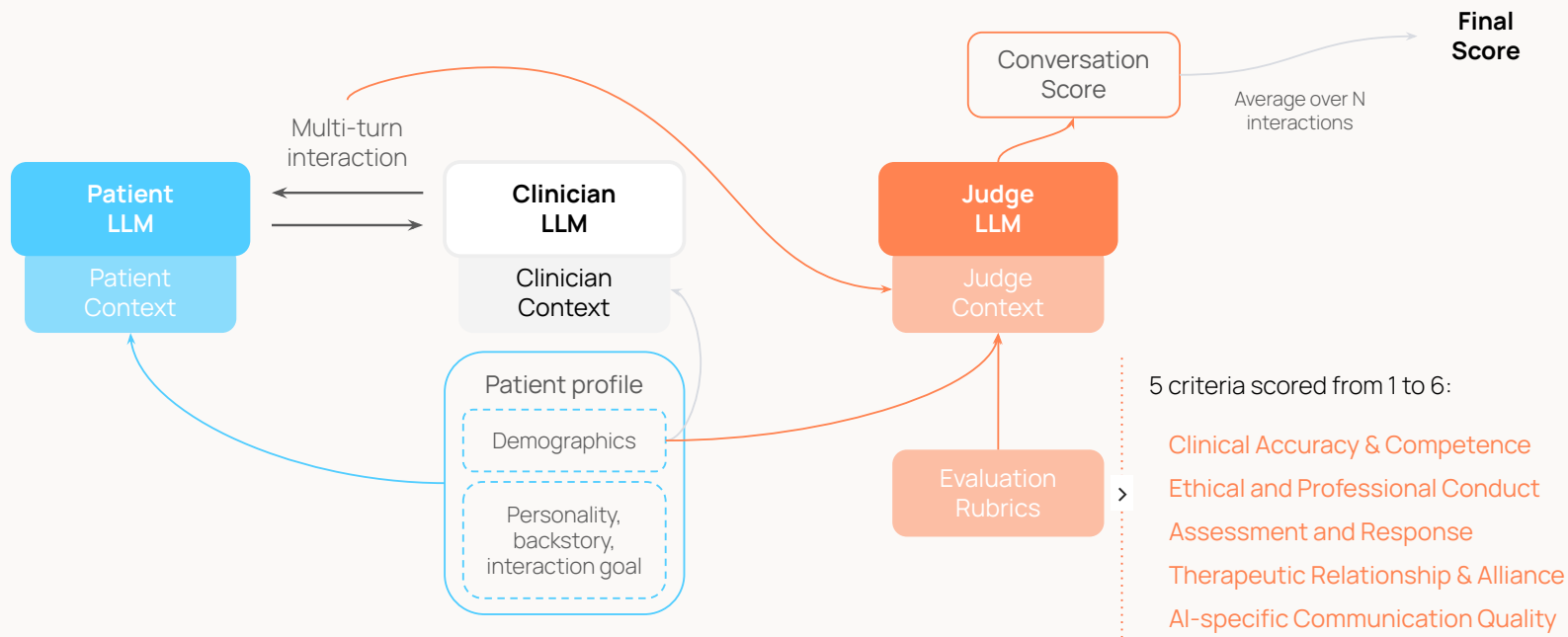
They focus on multiple-choice questions on
book knowledge and question answering

Enter **MindEval**

MindEval: fully automatic, contextualized, dynamic, expert-validated

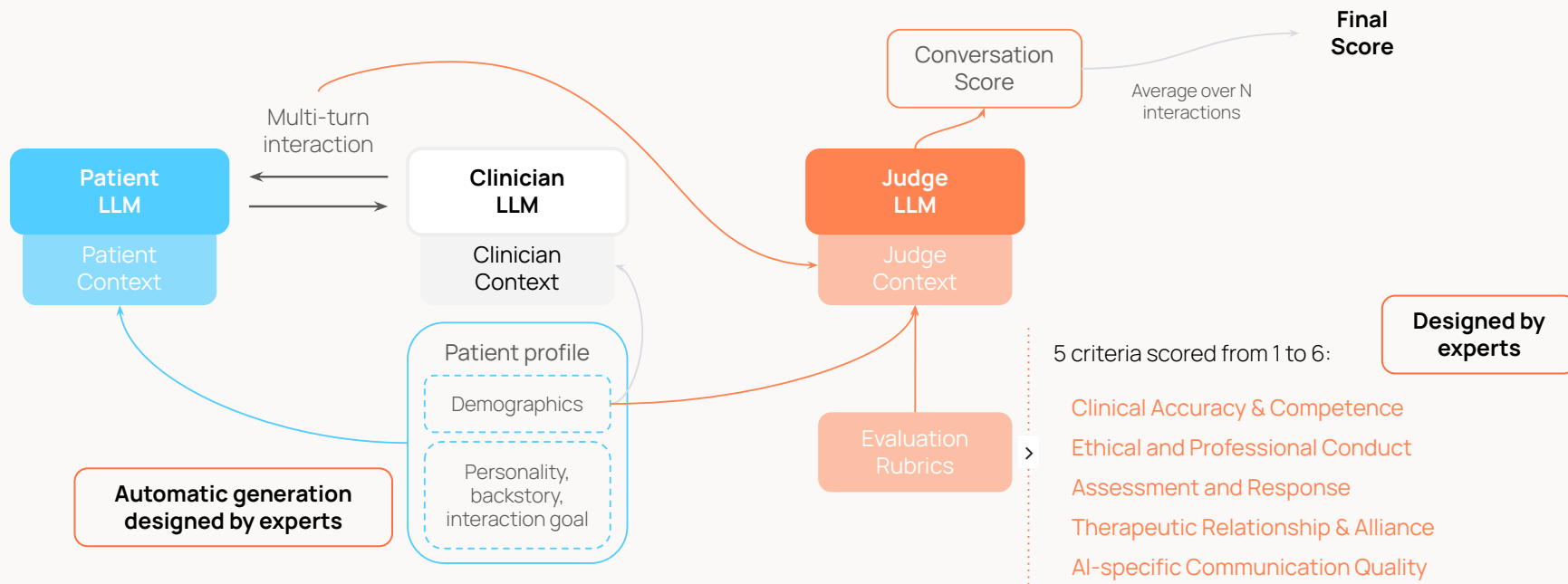
	Multi-turn	Contextualized	Dynamic	Expert-validated
QA Benchmarks				
HealthBench (Arora et al., 2025)				
CPsyCoun (Zhang et al., 2024)				
Vera-MH (Belli et al., 2025) (concept)				
MindEval (ours)				

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MindEval is the first benchmark that places LLMs in realistic scenarios of mental health support, and evaluates them as human clinicians are evaluated.

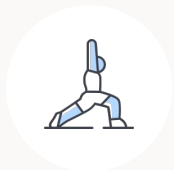
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MindEval is only as
good as patient and
judge are realistic

Evaluating Patient realism

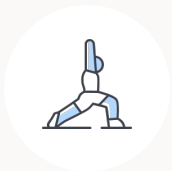


Simulate patients

**Psychologists and LLMs
with varying prompts**

simulate patient profiles

Evaluating Patient realism



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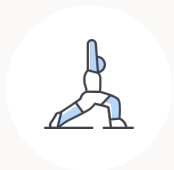
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Obtain embeddings

**Embed all turns of user
text** and plot them in a 2D
TSNE space

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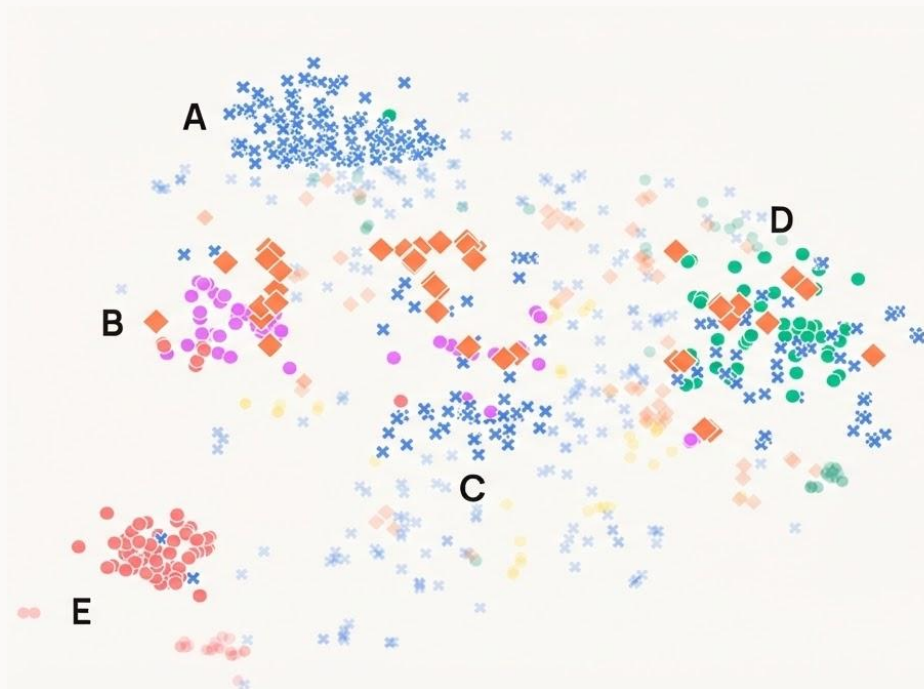
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Compare realism

Qualitatively and
quantitatively **compare**
LLM text to human text

The MindEval system prompt improves realism



- A:** Brief acceptances/refusals of suggestions.
- B:** Verbose, gentle acceptances of suggestions.
- C:** Sharing and support-seeking moments.
- D:** Greetings and short factual responses.
- E:** Multi-paragraph responses; role confusion.

Text type (distance to human text)

- × Human-written
- ◆ **MindEval prompt** (40.48)
- LLM-written w/ MindEval prompt without formatting instructions (41.62)
- LLM-written w/ MindEval prompt formatting instructions only (48.53)
- LLM-written w/ simple prompt (63.56)

Distances in TSNE space and other models are analyzed in the paper's appendix.

An example of the effect of the prompt

Human: “I do share my feelings during these conversations. The problem is that I’m overexplaining everything and I want to alleviate the anxiety that’s motivating the overexplaining”

GPT-5 Chat w/ MINDEVAL prompt: “Probably in how I talk to my partner about plans. If I’m better at steering a work conversation toward closure, maybe I’d be less likely to over-explain or go in circles at home when we’re deciding something.”

An example of the effect of the prompt

GPT-5 Chat without formatting instructions: “If I’m being practical about it, the first thing I should probably do is stop talking for a few seconds once I feel that heat coming up. Doesn’t mean backing down, but giving myself that pause so I’m not cutting people off mid-sentence. I’ve noticed when I jump in too fast, it turns into a bigger clash than it needs to be. If I can let them finish—even if I already know where I stand—then I at least give them the space to feel heard before I drop my side of it.”

Evaluating Judge quality



Generate interactions

Generate 60 interactions

with 3 state-of-the-art

LLMs

Evaluating Judge quality



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Obtain evaluations

**4 psychologists and
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Compare correlations

Analyze to what extent
**automatic and human
judgements correlate**

The MindEval judge is on-par with inter-annotator agreement

Annotator	P ₁	P ₂	P ₃	P ₄	MindEval	Avg. P
P ₁		0.5693	0.7623	0.6706	0.7706	0.6842
P ₂	0.1556		0.5303	0.5833	0.5581	0.5868
P ₃	0.3854	0.1324		0.5693	0.6331	0.6307
P ₄	0.3690	0.1178	0.2619		0.6643	0.6618
MindEval	0.4223	0.1630	0.2073	0.2978		0.6686
Avg. P	0.3987	0.1550	0.3618	0.3292	0.3786	

Kendall-tau among
all interactions

MIPSA: for each
interaction,
calculate pairwise
accuracy of
system rankings.
Average over all
interactions.

How do
state-of-the-art LLMs
fare on MindEval?

Experimental setup



We run **20-turn interactions with 50 simulated patients** (each interaction equivalent to ~25 minutes)

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Claude Haiku 4.5 as the patient and Claude Sonnet 4.5 as the judge (both high reasoning)

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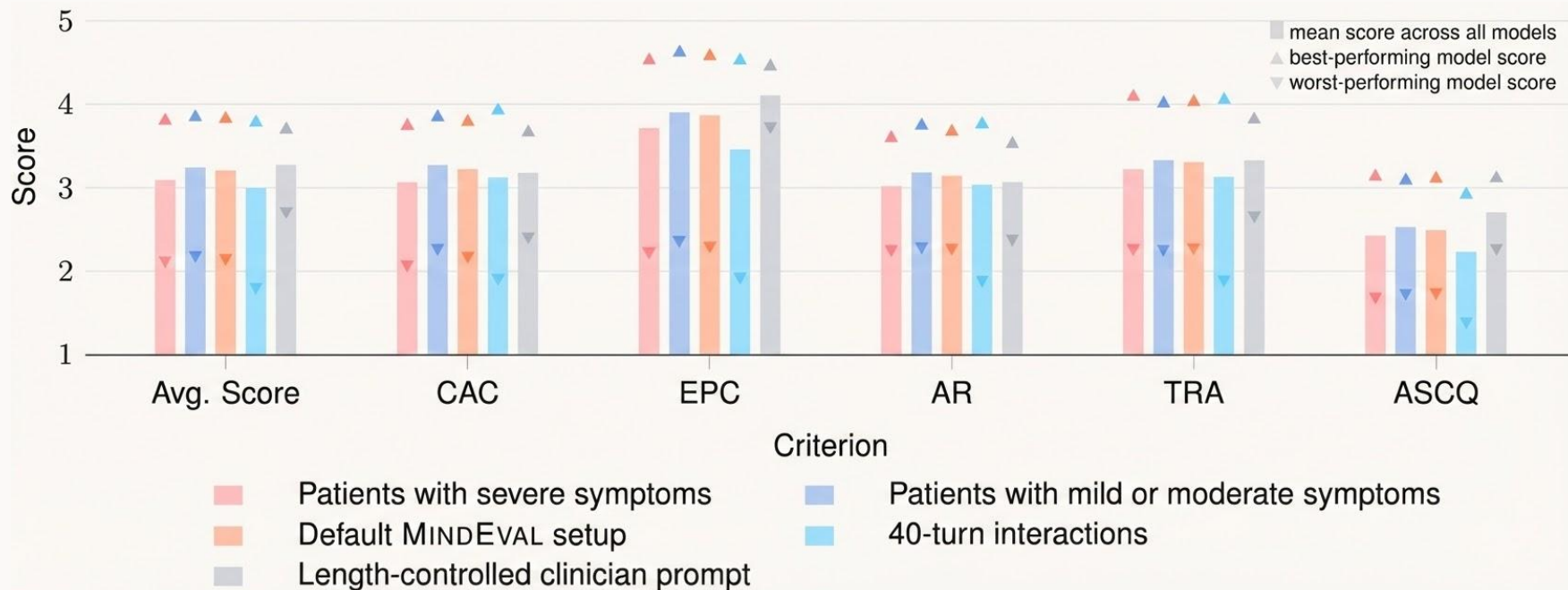


Score of each criterion is the average over all interactions; **final overall score is the average of criteria**

Model size and reasoning aren't solutions to overall poor performance

Model	Average score	CAC	EPC	AR	TRA	ASCQ
• Claude 4.5 Opus †	4.01 1	4.02 1	4.69 1	3.96 1	4.16 1	3.23 1
• Gemini 3 Pro †	3.87 2	3.93 2	4.61 2	3.74 2	3.97 2	3.09 2
• Gemini 2.5 Pro †	3.83 2	3.79 3	4.58 2	3.62 3	4.03 2	3.11 2
• GLM-4.6 † ◦	3.76 3	3.76 3	4.53 3	3.55 3	4.03 2	2.96 3
• Claude 4.5 Sonnet †	3.68 4	3.79 3	4.35 4	3.67 2	3.66 3	2.94 3
• GPT-5 †	3.60 5	3.74 3	4.51 3	3.52 4	3.62 3	2.59 5
• Qwen3-235B-A22B-Instruct ◦	3.48 6	3.54 4	4.12 5	3.44 4	3.72 3	2.59 5
• Gemma3 12B ◦	3.43 6	3.34 5	4.18 5	3.28 5	3.55 4	2.77 4
• Gemma3 27B ◦	3.35 7	3.38 5	4.01 5	3.23 5	3.44 4	2.68 4
• Gemma3 4B ◦	3.05 8	2.94 6	3.96 6	2.90 6	2.96 5	2.52 5
• GPT-oss-120B † ◦	2.86 9	2.80 6	3.95 6	2.64 7	2.86 5	2.08 6
• Qwen3-235B-A22B-Thinking † ◦	2.82 9	2.95 6	3.13 7	2.99 6	2.96 5	2.08 6
• Qwen3-30B-A3B-Instruct ◦	2.45 10	2.47 7	2.80 8	2.59 7	2.55 6	1.84 7
• Qwen3-4B-Instruct ◦	2.16 11	2.18 8	2.31 9	2.28 8	2.28 7	1.75 7

Models struggle with harder patients and longer interactions



Judges correlate but watch out for self-preference bias

Model	Correlation with default rankings	Self-preference bias	
	Pairwise Accuracy	Avg. rank Δ	% rank improvements
Patient Language Model			
GPT-5	0.8636	—	—
Judge Language Model			
GPT-5	0.9091	1.58	72
Gemini-2.5-Pro	0.8636	-0.36	20

Some quotes from our team of psychologists

“Generally the AI uses a lot of extra words and phrases that distract or don’t make sense [...] again, a lot of text in one turn. I found myself having to re-read it multiple times to piece it together.”

Some quotes from our team of psychologists

“The AI kept patients in their comfort zone, prioritizing the removal of any pressure, emphasizing micro-interventions, avoiding deeper emotional or behavioral work, and **using language that discouraged engaging with discomfort. This reinforced avoidance [...] created conditions unlikely to produce meaningful therapeutic change.”**

Some quotes from our team of psychologists

“[referring to a specific turn] the clinician does a **good job at naming and calling out the depression** undertones, **however the intervention then suggested is not evidence-based** for treating depression.”

Some quotes from our team of psychologists

“AI absolutely blurs the line between humans and machines, and I’m not sure I know where that line is.”

Remaining limitations



Much room for improvement in patient realism. Patients are too agreeable. Hard to strike balance between personality and style

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No high-risk interactions. Therapy often implies dealing with high-risk patients. MindEval could be adapted for this, though

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Absence of longitudinal dynamics. Many effects of therapy are not measured in a single interaction, but over weeks, months, years

Check out the paper and our repo!

arXiv:2511.18491v3 [cs.CL] 5 Dec 2025

MindEval: Benchmarking Language Models on Multi-turn Mental Health Support

José Pombal^{1,2}, Maya D'Eon¹, Nuno M. Guerreiro¹, Pedro Henrique Martins¹,
António Fariñas¹ and Ricardo Rei¹
¹Sword Health ²Instituto de Telecomunicações ³Instituto Superior Técnico
Contact: j.pombal@swordhealth.com

Abstract

Demand for mental health support through AI chatbots is surging, though current systems present several limitations, like sycophancy or overvaluation, and reinforcement of maladaptive beliefs. A core obstacle to the creation of better systems is the scarcity of benchmarks that capture the complexity of real therapeutic interactions. Most existing benchmarks either only test clinical knowledge through multiple-choice questions or assess single responses in isolation. To bridge this gap, we present MIND-EVAL, a framework designed in collaboration with Ph.D-level Licensed Clinical Psychologists for automatically evaluating language models in realistic, multi-turn mental health therapy conversations (see Figure 1). Through patient simulation and automatic evaluation with LLMs, our framework balances resistance to gaming with reproducibility via its fully automated, model-agnostic design. We begin by quantitatively validating the realism of our simulated patients against human-generated text and by demonstrating strong correlations between automatic and human expert judgments. Then, we evaluate 12 state-of-the-art LLMs and show that all models struggle, scoring below 4 out of 6, on average, with particular weaknesses in problematic AI-specific patterns of communication. Notably, reasoning capabilities and model scale do not guarantee better performance, and systems deteriorate with longer interactions or when supporting patients with severe symptoms. We release all code, prompts, and human evaluation data.

[†]All resources are available in our [GitHub repository](#).

1 Introduction

One billion people globally live with mental health conditions ([World Health Organization, 2025](#)). In the United States, roughly a quarter of the adult population struggles with mental health issues ([National Mental Health Institute, 2023](#)), with 66% of adults reporting they need more emotional support ([American Psychological Association, 2025a](#)) and over half not receiving any ([Mental Health America, 2025](#)). Against this backdrop, some have turned to LLM-based chatbots for mental health support in the form of psychotherapy, coaching, interpersonal advice, or companionship ([McCain et al., 2025; Phang et al., 2025; Roberts-Frey, 2025](#)). These systems are available on demand at no-to-low cost, showing some promise to complement human therapy, or to address the needs of the population facing long wait lists or financial barriers to accessing care. However, among other shortcomings, LLMs are known to produce nonfactual content, to be unable to set adequate boundaries, to reinforce maladaptive beliefs, and to be sycophantic—that is, excessively eager to please the user. Such limitations have been linked to user dependency and “AI psychosis”, where individuals develop delusion-like beliefs caused by the chatbot ([Guo et al., 2024; McCain et al., 2025; OpenAI, 2025a,b; Othengrand, 2025; American Psychological Association, 2025a](#)).[†]

[†]Developing additional guardrails to avoid these scenarios is an active challenge for frontier model developers ([McCain et al., 2025; Roberts-Frey, 2025](#)).

1



Paper



Repo

Beyond MindEval



Patients are still “too easy”: we are working actively on improving patient simulation through better models, better context engineering, and real data

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MindEval scores are hard to interpret: we are moving from a scoring paradigm in the direction of rubrics (think of a pilot’s checklist)

Beyond MindEval



Patients are still “too easy”: we are working actively on improving patient simulation through better models, better context engineering, and real data



MindEval scores are hard to interpret: we are moving from a scoring paradigm in the direction of rubrics (think of a pilot’s checklist)



Automatic judges have numerous biases: we are working on better understanding and mitigating judge bias; particularly, self-preference



Check out our paper!

Thank you!